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Lab 4

Main.c



QEMU - Press Ctrl+Alt+G to release grab

Machine View

GNU nano 6.2

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <stdlib.h>
#include <unistd.h>
#define MAXN 50
int n;
pthread_mutex_t forks[MAXN];
sem_t plates;
void* philosopher(void* id) {
    int i = *(int*)id;
    while (1) {
        printf("Philosopher %d is thinking\n", i);
        sleep(rand() % 3);
        sem_wait(&plates);
        pthread_mutex_lock(&forks[i]);
        pthread_mutex_lock(&forks[(i+1) % n]);
        printf("Philosopher %d is eating\n", i);
        sleep(rand() % 2);
        pthread_mutex_unlock(&forks[i]);
        pthread_mutex_unlock(&forks[(i + 1) % n]);
        sem_post(&plates);
    }
}

int main() {
    printf("Enter number of philosopher (>3): ");
    if (scanf("%d", &n) != 1) {
        printf("Input error!\n");
        return 1;
    }
    if (n <= 3 || n > MAXN) {
        printf("Error: N must be > 3 and <= %d\n", MAXN);
        return 1;
    }
    pthread_t tid[n];
    int id[n];
    sem_init(&plates, 0, 2);
    for (int i=0; i<n; i++)
        pthread_mutex_init(&forks[i], NULL);
    for (int i=0; i<n; i++) {
        id[i] = i;
        pthread_create(&tid[i], NULL, philosopher, &id[i]);
    }
    for (int i=0; i<n; i++)
        pthread_join(tid[i], NULL);
    return 0;
}
```

Result:



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Machine View

```
Philosopher 0 is thinking
Philosopher 3 is thinking
Philosopher 1 is eating
Philosopher 2 is eating
Philosopher 1 is thinking
Philosopher 3 is eating
Philosopher 2 is thinking
Philosopher 0 is eating
Philosopher 3 is thinking
Philosopher 1 is eating
Philosopher 0 is thinking
Philosopher 1 is thinking
Philosopher 2 is eating
Philosopher 0 is eating
Philosopher 2 is thinking
Philosopher 2 is eating
Philosopher 2 is thinking
Philosopher 0 is thinking
Philosopher 1 is eating
Philosopher 3 is eating
Philosopher 3 is thinking
Philosopher 1 is thinking
Philosopher 0 is eating
Philosopher 2 is eating
Philosopher 2 is thinking
Philosopher 3 is eating
Philosopher 0 is thinking
Philosopher 2 is eating
Philosopher 3 is thinking
Philosopher 1 is eating
Philosopher 2 is thinking
Philosopher 0 is eating
Philosopher 1 is thinking
Philosopher 3 is eating
Philosopher 3 is thinking
Philosopher 0 is thinking
Philosopher 2 is eating
Philosopher 2 is thinking
Philosopher 2 is eating
Philosopher 3 is eating
Philosopher 2 is thinking
Philosopher 3 is thinking
Philosopher 1 is eating
Philosopher 0 is eating
Philosopher 1 is thinking
Philosopher 0 is thinking
Philosopher 3 is eating
Philosopher 2 is eating
Philosopher 3 is thinking
```